

Lancaster University College
at Beijing Jiaotong University

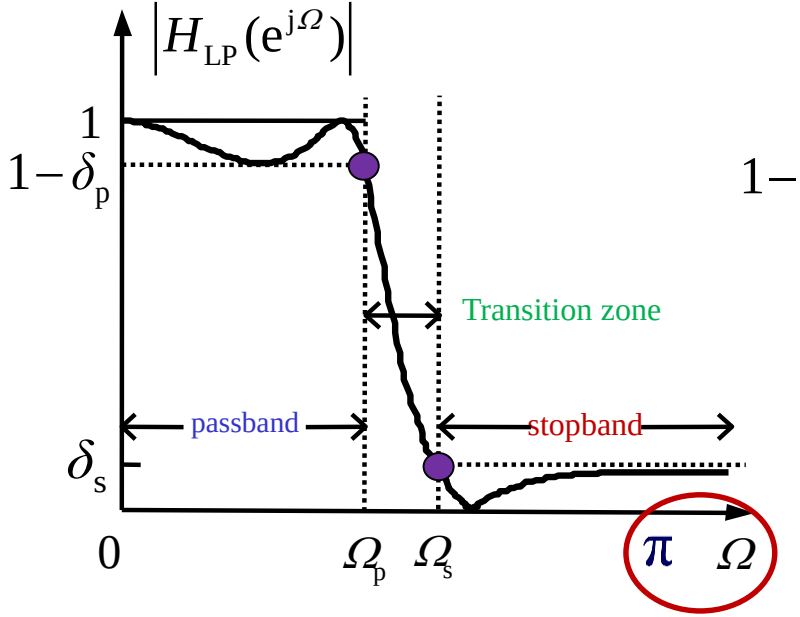
Digital Signal Processing

Zhou Hang

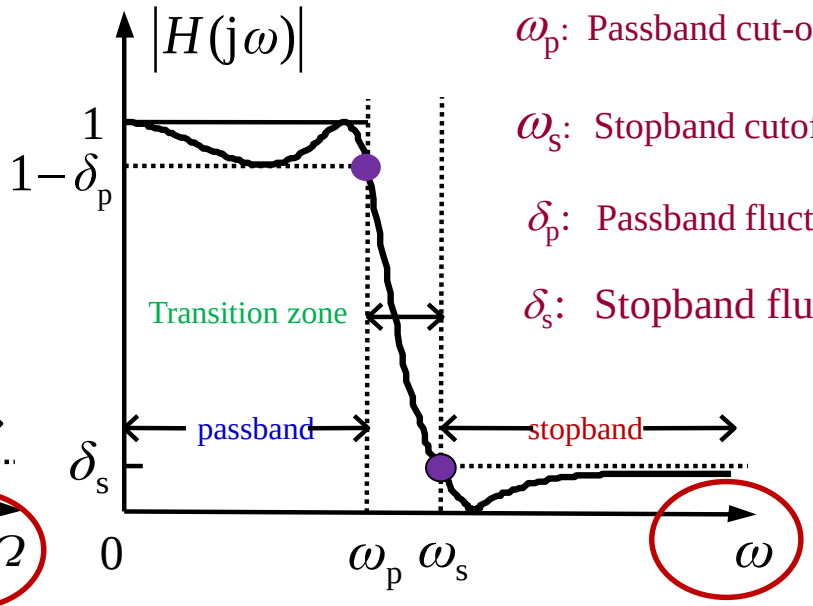
Analog low-pass filter design

- Technical specifications for analog filters
- Butterworth analog low-pass filter
- Chebyshev Type I analog low-pass filter
- Chebyshev Type II analog low-pass filter
- Elliptic analog low-pass filter

Technical specifications for analog filters



Low pass digital filter



Low-pass analog filter

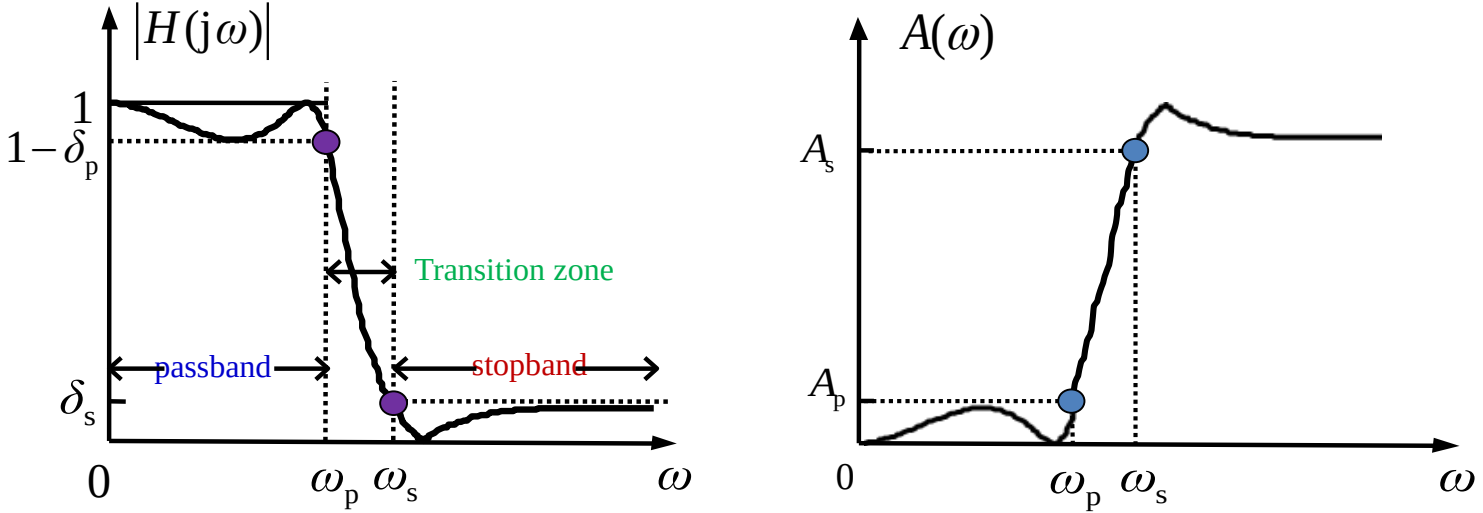
ω_p: Passband cut-off frequency

ω_s: Stopband cutoff frequency

δ_p: Passband fluctuation

δ_s: Stopband fluctuation

Technical specifications for analog filters

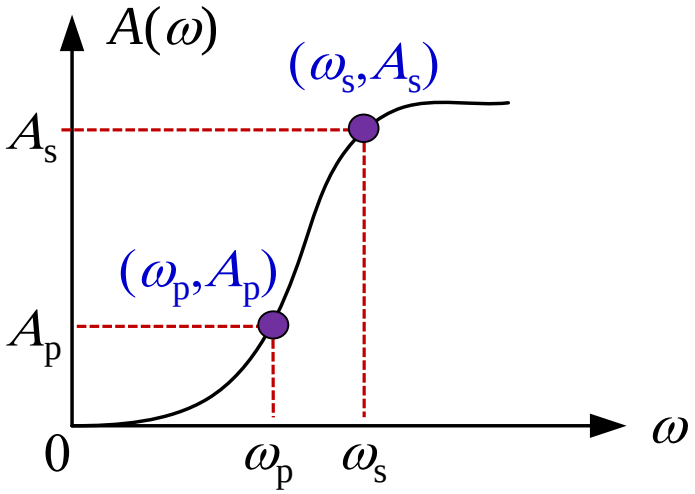


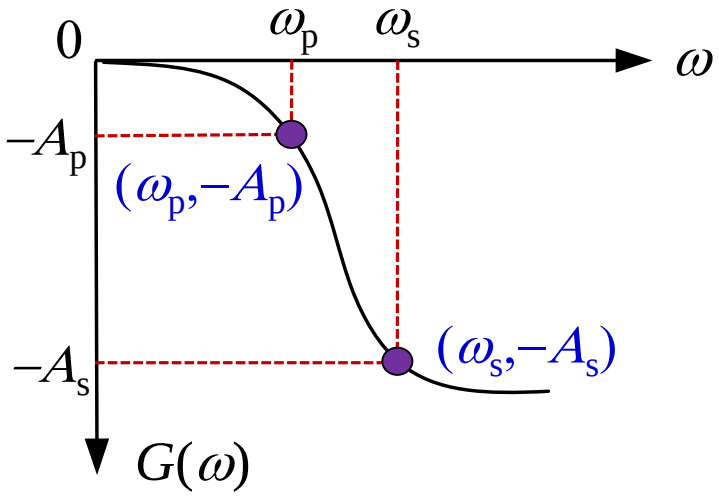
Attenuation function(dB) $A(\omega) = -10\lg|H(j\omega)|^2 = -20\lg|H(j\omega)|$

Maximum passband attenuation(dB) $A_p = -10\lg(1-\delta_p)^2 = -20\lg(1-\delta_p)$

Stopband minimum attenuation (dB) $A_s = -10\lg\delta_s^2 = -20\lg\delta_s$

Technical specifications for analog filters

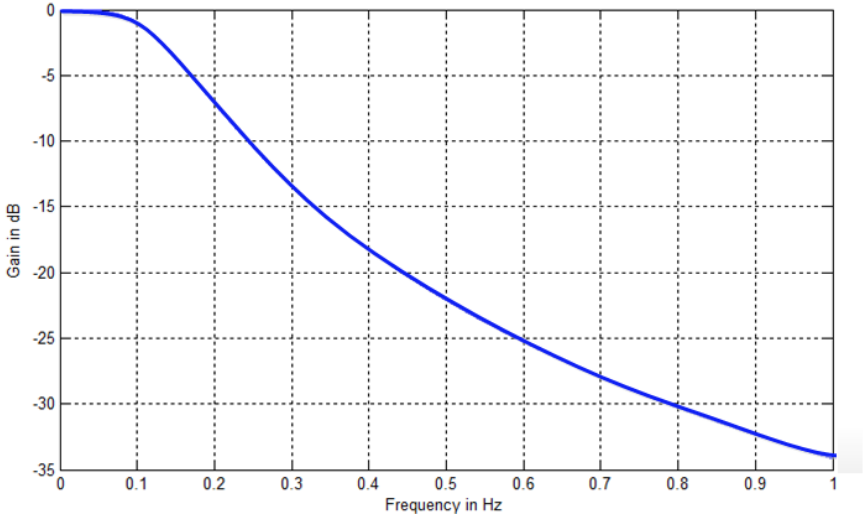




Attenuation function $A(\omega) = -20\lg\left|H(\mathrm{j}\omega)\right|$

Gain function $G(\omega) = -A(\omega)$

Technical specifications for analog filters



Gain function $G(\omega)$

Butterworth Analog low-pass filter

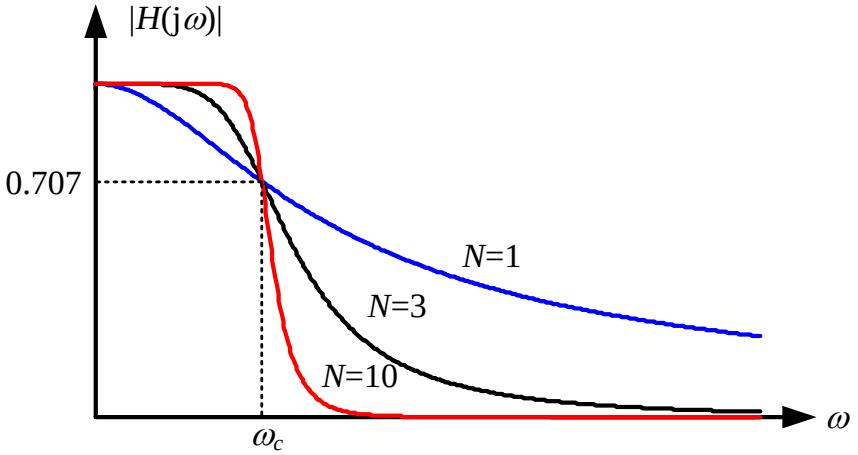
BW Frequency-domain characteristics of analog low-pass filters

$$|H(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2N}}$$

N: Filter order

ω_c: 3dB Cut-off frequency

$A(\omega_c) = -10\lg|H(j\omega_c)|^2 = 10\lg 2 \approx 3\text{dB}$



Amplitude response of a typical BW low-pass filter

Butterworth Analog low-pass filter

BWSimulation of frequency domain characteristics of low-pass filters

$$|H(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2N}}$$

(1) $|H(j0)|=1$, $|H(j\infty)|=0$,

$A(\omega_c)=-10\lg|H(j\omega_c)|^2 = 10\lg 2 \approx 3\text{dB}$,

So w_c is called 3dB cut frequency. If w_c =1, it is normalized BWF.

w_c is generally not a passband truncated w_p

(2) Amplitude response monotonically decreases

(3) $|H(jw)|^2$ the derivative from 1 to $2N$-1 is zero at w=0, which is said to have maximum flatness at w=0.

Butterworth Analog low-pass filter

- Butterworth filters were first proposed by British engineer Stephen Butterworth in 1930 in a paper titled "On the Theory of Filter Amplifiers" published in the British journal of Radio Engineering.
- The Butterworth analog Filter has the smoothest frequency response curve at the passband (Maximally Flat Magnitude Filter).

Butterworth Analog low-pass filter

Design steps of BW analog low-pass filter $|H(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2N}}$

Step 1: Determine the order N of the analog filter

Step 2: Determine the 3dB cut frequency wc of the analog filter

Step 3: Calculate the poles of the analog filter's system function

Step 4: Get the system function $H_L(s)$ for the analog low-pass filter

Butterworth Analog low-pass filter

Design steps of BW analog low-pass filter

$$|H(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2N}}$$

Step 1: Determine the order N of the analog filter

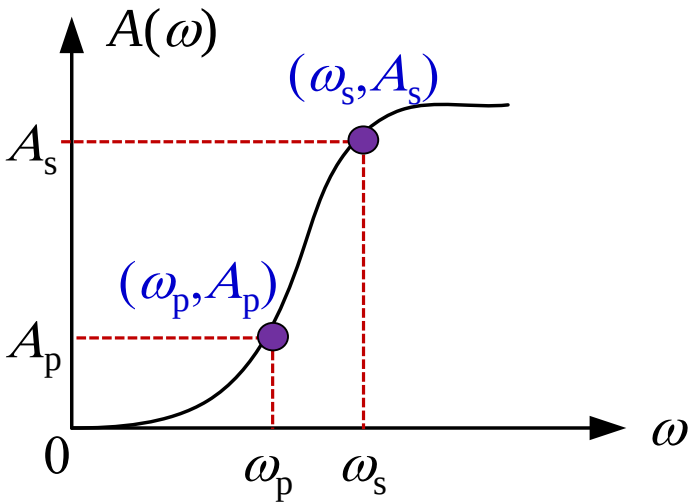
Given conditions (ω_p, A_p), (ω_s, A_s)

On the basis of:

$$A(\omega) = -10\lg|H(j\omega)|^2 = 10\lg[1+(\omega/\omega_c)^{2N}]$$

Plug in the known conditions to get:

$$N \geq \frac{\lg\left(\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\omega_s/\omega_p)}$$

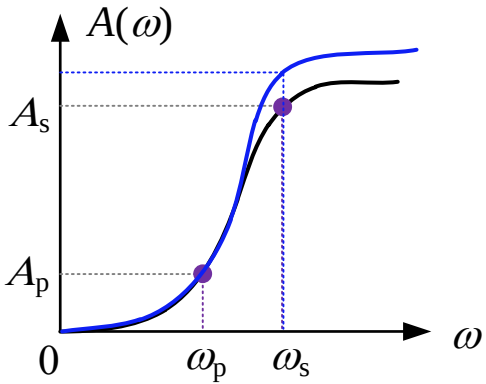


Butterworth Analog low-pass filter

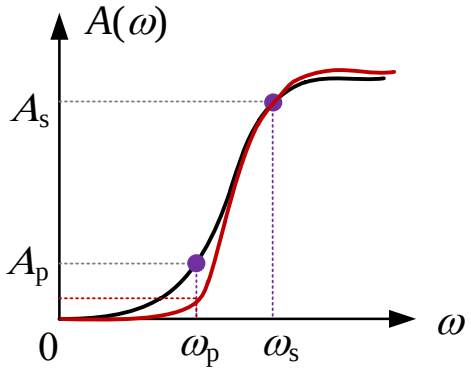
Design steps of BW analog low-pass filter

Step 2: Determine the 3dB cut frequency wc of the analog filter

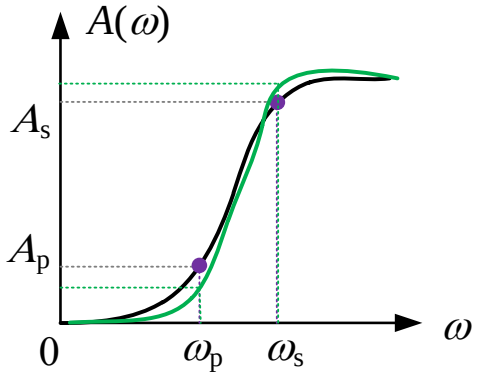
$$\frac{\omega_p}{(10^{0.1A_p}-1)^{\frac{1}{2N}}} \le \omega_c \le \frac{\omega_s}{(10^{0.1A_s}-1)^{\frac{1}{2N}}}$$



ω_c is determined by the passband boundary point ω_p



ω_c is determined by the stopband boundary point ω_s



ω_c is determined by the transition zone area points

Butterworth Analog low-pass filter

Design steps of BW analog low-pass filter

Step 3: Determine the system function poles of the analog filter

Using real coefficients to simulate the conjugate symmetry of the frequency response of the system,

$$\left|H(\mathrm{j}\omega)\right|^2 = H(\mathrm{j}\omega)H^*(\mathrm{j}\omega) = H(\mathrm{j}\omega)H(-\mathrm{j}\omega) = H(s)H(-s)\big|_{s=\mathrm{j}\omega}$$

Find the poles of the system function as follows:

$$s_k = \omega_c \mathrm{e}^{\mathrm{j}\pi\left(\frac{1}{2}+\frac{2k-1}{2N}\right)}; \quad k = 1, 2, \ldots, 2N$$

Butterworth Analog low-pass filter

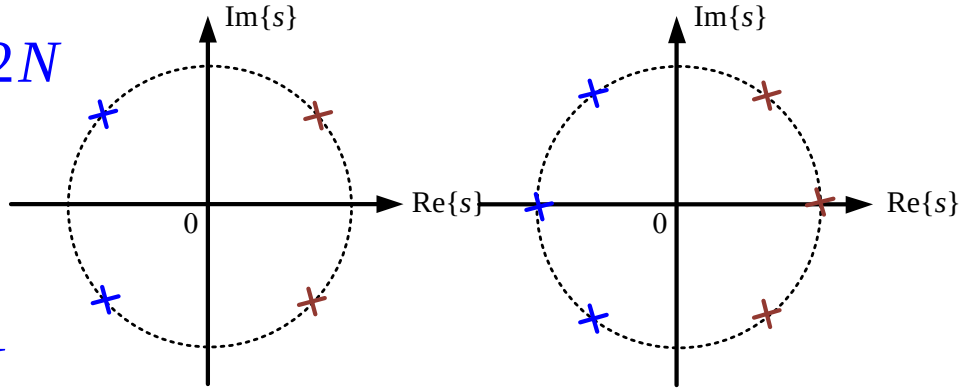
Design steps of BW analog low-pass filter

Step 3: Determine the system function poles of the analog filter

$$s_k = \omega_c e^{j\pi(\frac{1}{2}+\frac{2k-1}{2N})}; \quad k = 1, 2, ..., 2N$$

Take the poles in the left half plane of s:

$$s_k = \omega_c e^{j\pi(\frac{1}{2}+\frac{2k-1}{2N})}; \quad k = 1, 2, ..., N$$



N is even (N=2)　　N is odd (N=3)

Butterworth Analog low-pass filter

Design steps of BW low-pass analog filter

Step 4: Determine the system function H_L (s) for the analog low-pass filter

$$H_{\mathrm{L}}(s)=\prod_{k=1}^{N}\frac{-s_k}{s-s_k} \qquad s_k=\omega_{\mathrm{c}}\mathrm{e}^{\mathrm{j}\pi\left(\frac{1}{2}+\frac{2k-1}{2N}\right)};\quad k=1,2,\ldots,N$$

[case] A BW type analog low-pass filter is designed to meet the following criteria ω_p=0.1π rad/s, ω_s=0.4π rad/s, A_p≤1dB, A_s≥10dB。

Solution: (1) Calculate the order N of the analog low-pass filter and the 3dB cut frequency ω_c

$$N \geq \frac{\lg\left(\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg(\omega_s/\omega_p)} = 1.28$$

Take N=2 and bring N=2 into the equation that satisfies the passband boundary frequency ω_p

$$\omega_c = \frac{\omega_p}{(10^{0.1}-1)^{1/4}} = 0.4404 \text{ rad/s}$$

[case] A BW type analog low-pass filter is designed to meet the following criteria ω_p=0.1π rad/s, ω_s=0.4π rad/s, A_p≤1dB, A_s≥10dB。

Solution:(2) Determine the poles of the analog low-pass filter

$$s_k = \omega_c \mathrm{e}^{\mathrm{j}\pi(\frac{1}{2}+\frac{2k-1}{2N})};\quad k=1,2,...,N$$

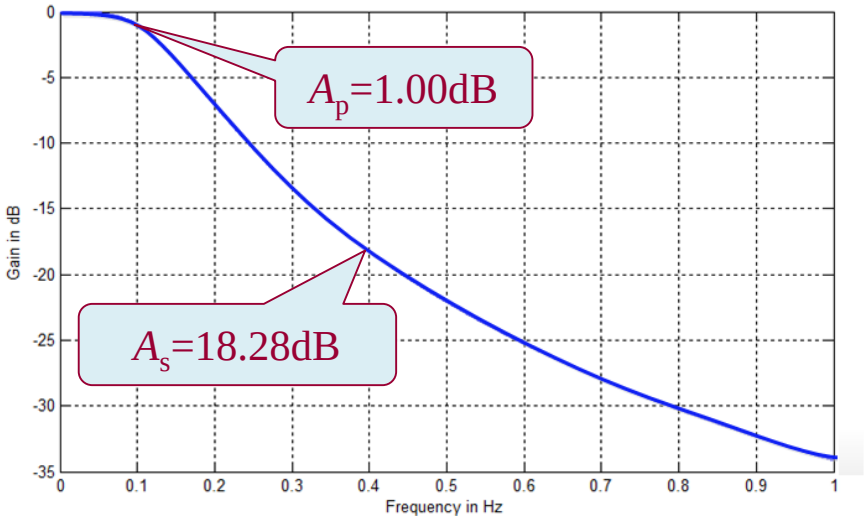
$$s_1 = \omega_c \mathrm{e}^{\mathrm{j}\frac{3}{4}\pi} = \omega_c(-\frac{\sqrt{2}}{2}+\mathrm{j}\frac{\sqrt{2}}{2}) \qquad s_2 = \omega_c \mathrm{e}^{\mathrm{j}\frac{5}{4}\pi} = \omega_c(-\frac{\sqrt{2}}{2}-\mathrm{j}\frac{\sqrt{2}}{2})$$

(3) Determine the system function H_L (s) of the analog low-pass filter

$$H_L(s) = \prod_{k=1}^{N}\frac{-s_k}{s-s_k} = \frac{\omega_c^2}{s^2+\sqrt{2}\omega_c s+\omega_c^2} = \frac{0.4404^2}{s^2+0.6228s+0.4404^2}$$

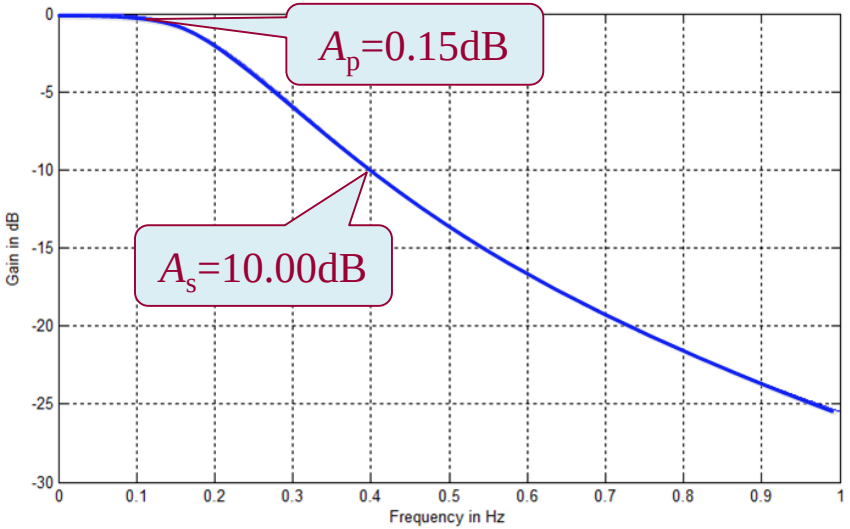
[case] A BW type analog low-pass filter is designed to meet the following criteria

ω_p=0.1π rad/s, ω_s=0.4π rad/s, A_p≤1dB, A_s≥10dB。



When wc is determined by the passband boundary point ω_p:

A_p meets the target. A_s there is a margin.



When wc is determined by the stopband boundary point ω_s:

A_p exists margin; A_s meets the indicator.

Chebyshev Type I analog low-pass filter



Chebyshev (Chebyshev), formerly known as Russian П а ф н́ т и й Л ь kind guide о́ kind guide и discusses some related problems Ч е б ы ш ё kind guide, (on May 26, 1821 - December 8, 1894), the Russian mathematician. During his lifetime, he published more than 70 scientific papers on number theory, probability theory, function approximation theory, integral theory and so on. He proved Beltran's formula, the theorem on the distribution of prime numbers in natural series, the general formula for the law of large numbers, and the central limit theorem. He attached great importance not only to pure mathematics, but also to its application.

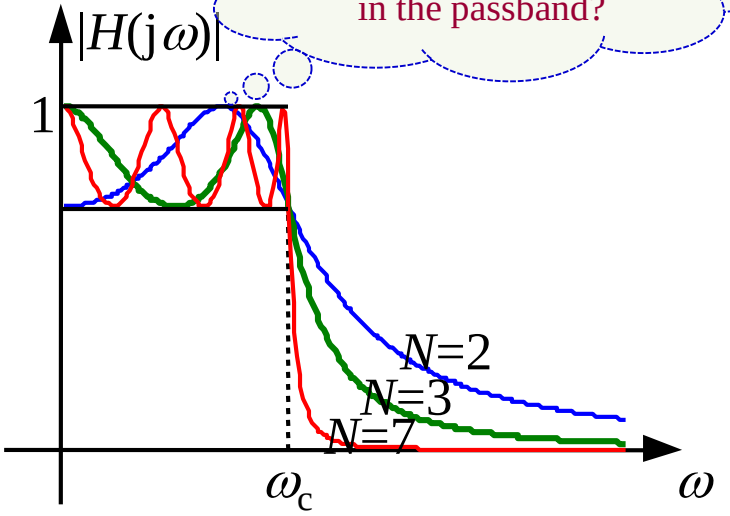
Chebyshev Type I analog low-pass filter

Frequency domain characteristics of CB-I low-pass analog filter

$$\left|H(\mathrm{j}\omega)\right|^2 = \frac{1}{1+\varepsilon^2 C_N^2(\omega/\omega_c)}$$

$$C_N(x) = \begin{cases} \cos[N\arccos(x)] & |x| \le 1 \\ \cosh[N\,\mathrm{arccos\,h}(x)\,] & |x| > 1 \end{cases}$$

Thinking: What are the benefits of CB I ripples in the passband?



Amplitude response of CB-I low-pass filter

Chebyshev Type I analog low-pass filter

Frequency domain characteristics of CB-I low-pass analog filter

$$|H(\mathrm{j}\omega)|^2 = \frac{1}{1+\varepsilon^2 C_N^2(\omega/\omega_c)}$$

(1) $0 \le \omega \le \omega_c$, $|H(\mathrm{j}w)|^2$ in 1and $\dfrac{1}{1+\varepsilon^2}$ Equiamplitude oscillation

(2) $\omega > \omega_c$, $|H(\mathrm{j}\omega)|^2$ Monotone decline (the larger N, the faster the decline)

(3) $|H(\mathrm{j}\omega)|^2$ in ω=0 is

$$|H(\mathrm{j}0)|^2 = \begin{cases} 1, & N\text{为奇数} \\ 1/1+\varepsilon^2, & N\text{为偶数} \end{cases}$$

Chebyshev Type I analog low-pass filter

The design steps of CB-I low-pass analog filter $|H(\mathrm{j}\omega)|^2 = \dfrac{1}{1+\varepsilon^2 C_N^2(\omega/\omega_c)}$

Step 1: ω_c is determined by passband truncation ω_p

$$\omega_c = \omega_p$$

Step 2: Determine ε from passband attenuation A_p $\quad \varepsilon = \sqrt{10^{0.1A_p} - 1}$

Step 3: Determine N from the stopband indicator, ε, and ω_c

$$N \geq \frac{\operatorname{arccosh}\left(\frac{1}{\varepsilon}\sqrt{10^{0.1A_s} - 1}\right)}{\operatorname{arccosh}(\omega_s/\omega_p)}$$

Chebyshev Type I analog low-pass filter

The design steps of CB-I low-pass analog filter

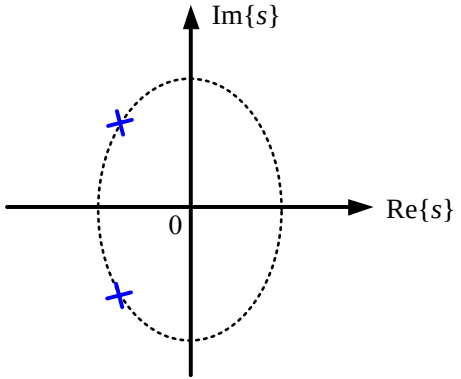
Step 4: Find the pole of the analog low-pass filter from the mode square $|H(\mathrm{j}\omega)|^2$

$$s_k = \sigma_k + \mathrm{j}\omega_k, k = 1,2,\cdots,N$$

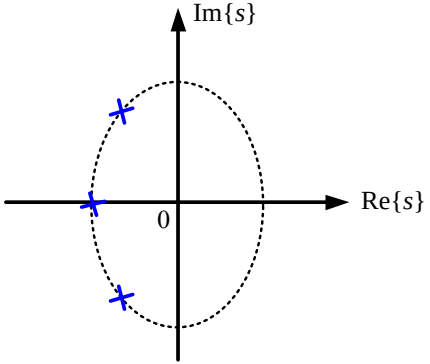
$$\sigma_k = -\sinh(\beta)\sin\frac{(2k-1)\pi}{2N}$$

$$\omega_k = -\cosh(\beta)\cos\frac{(2k-1)\pi}{2N}$$

其中 $\beta = \operatorname{arcsin}\mathrm{h}(1/\varepsilon)/N$



N=2



N=3

Chebyshev Type I analog low-pass filter

The design steps of CB-I low-pass analog filter

Step 5: Determine the system function $H(s)$ from the poles

$$H_{\mathrm{L}}(s)=\prod_{k=1}^{N}\frac{H_0}{s-s_k}$$

$$s_k=\sigma_k+\mathrm{j}\omega_k,\ k=1,2,\cdots,N$$

$$\sigma_k=-\sinh(\beta)\sin\frac{(2k-1)\pi}{2N}$$

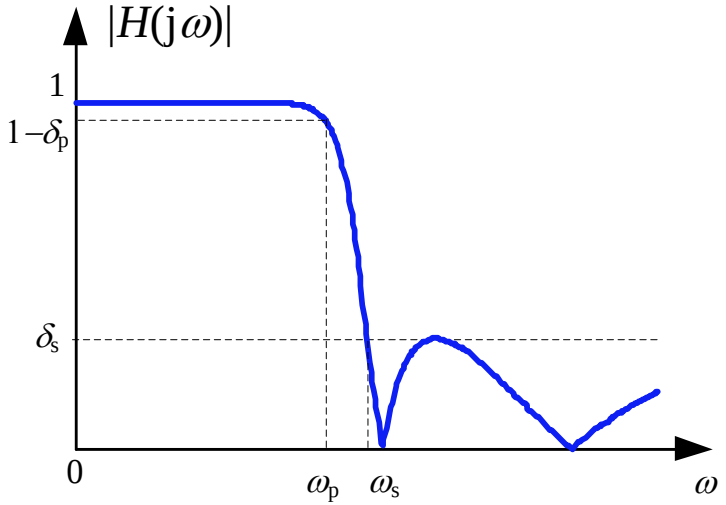
$$\omega_k=-\cosh(\beta)\cos\frac{(2k-1)\pi}{2N}$$

Chebyshev Type II analog low-pass filter

Frequency domain characteristics of CB II low-pass analog filter

$$|H(\mathrm{j}\omega)|^2 = 1 - \frac{1}{1+\varepsilon^2 C_N^2(\omega_c/\omega)} = \frac{\varepsilon^2 C_N^2(\omega_c/\omega)}{1+\varepsilon^2 C_N^2(\omega_c/\omega)}$$

ω_c: stopband truncation



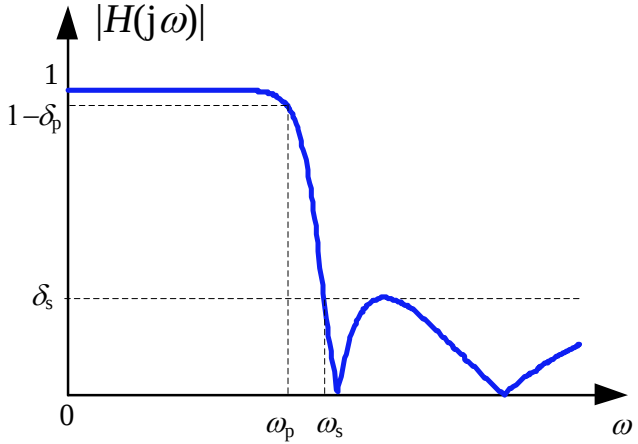
Amplitude response of CB II low-pass filter

Chebyshev Type II analog low-pass filter

Frequency domain characteristics of CB II low-pass analog filter

(1) in $|\omega| > \omega_c$, $0 \le |H(j\omega)|^2 \le \dfrac{\varepsilon^2}{1+\varepsilon^2}$

(2) To any N, ω_c and $\varepsilon > 0$, $|H(0)| = 1$



(3) In passband $0 \le \omega \le \omega_c$, $|H(j\omega)|^2$ monotone decline。

Chebyshev Type II analog low-pass filter

The design steps of CB II low-pass analog filter

Step 1: ω_c is determined by stopband truncation ω_s $\quad \omega_c = \omega_s$

Step 2: Determine e from stopband attenuation A_s $\quad \varepsilon = 1/\sqrt{10^{0.1A_s} - 1}$

Step 3: N is determined by the passband indicator, ε, and ω_c

$$N \geq \frac{\operatorname{arccosh}\left(\frac{1}{\varepsilon\sqrt{10^{0.1A_p} - 1}}\right)}{\operatorname{arccosh}(\omega_s / \omega_p)}$$

Chebyshev Type II analog low-pass filter

The design steps of CB II low-pass analog filter

Step 4: Find the pole of the analog low-pass filter from the mode square $|H(\mathrm{j}\omega)|^2$.

$$s_k = \sigma_k + \mathrm{j}\omega_k, k = 1,2,\cdots,N$$

Step 5: The system function is determined by the poles $H(s)$。

$$H_{\mathrm{L}}(s) = \prod_{k=1}^{N} \frac{H_0}{s - s_k}$$

* Elliptic analog low-pass filter

Frequency domain characteristics of elliptic analog low-pass filters

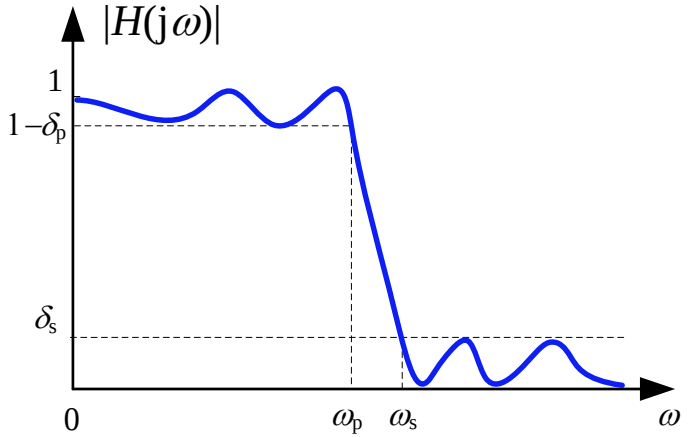
$$\left|H(\mathrm{j}\omega)\right|^2 = \frac{1}{1+\varepsilon^2 R_N^2(\omega/\omega_c)}$$

$R_N(x)$ is an elliptic function of order N with two parameters k and $k1$

(1) $|\omega|<\omega_p$时, $\dfrac{1}{1+\varepsilon^2} \le \left|H(\mathrm{j}\omega)\right|^2 \le 1$

(2) $\omega_c/k < |\omega| < \infty$ 时, $0 \le \left|H(\mathrm{j}\omega)\right|^2 \le \dfrac{1}{1+(\varepsilon/k_1)^2}$

(3) $\left|H(\mathrm{j}0)\right|^2 = \begin{cases} 1/(1+\varepsilon^2), & N\text{is even} \\ 1, & N\text{is odd} \end{cases}$



Amplitude response of elliptic low-pass filter

* Elliptic analog low-pass filter

Design steps of elliptic analog low-pass filter

Step 1: ω_c is determined by passband truncation ω_p

$$\omega_c = \omega_p$$

Step 2: Determine e from passband attenuation A_p

$$\varepsilon = 1 \Big/ \sqrt{10^{0.1A_p} - 1}$$

Step 3: Determine k by passband and stopband truncation

$$k = \omega_p / \omega_s$$

* Elliptic analog low-pass filter

Design steps of elliptic analog low-pass filter

Step 4: k_1 is determined by stopband attenuation A_s

$$k_1 = \varepsilon / \sqrt{10^{0.1A_s} - 1}$$

Step 5: Determine the order N

$$N = \frac{K(k)K(\sqrt{1-k_1^2})}{K(\sqrt{1-k^2})K(k_1)}$$

Step 6: Determine the system function $H_L(s)$ based on the above parameters

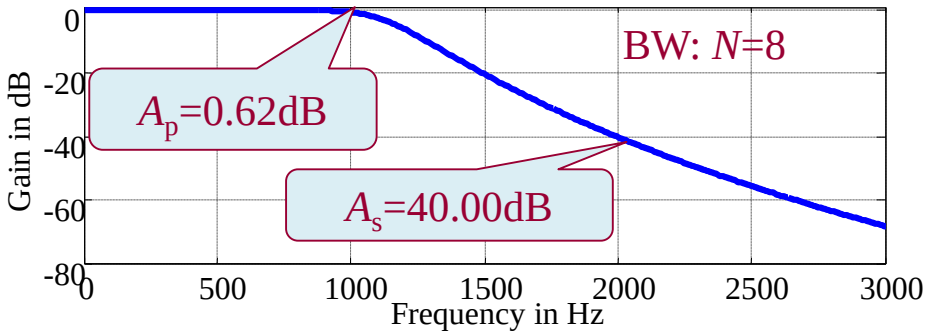
Comparison of BW, CB and elliptic analog low-pass filters

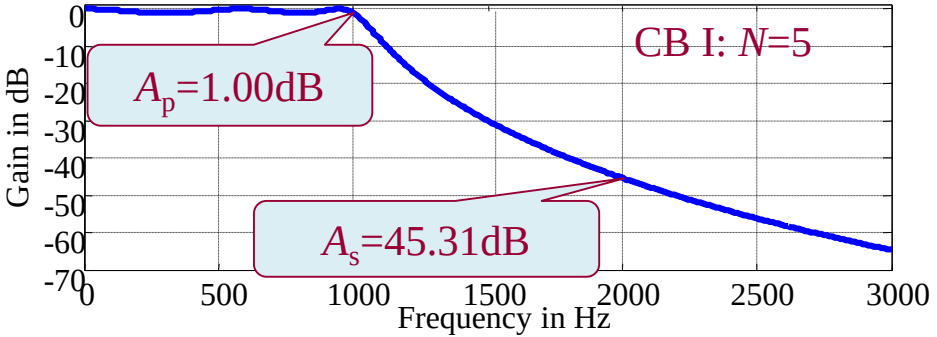
A BW type, CB-I type and elliptical type analog low-pass filter is designed to meet the following technical specifications：

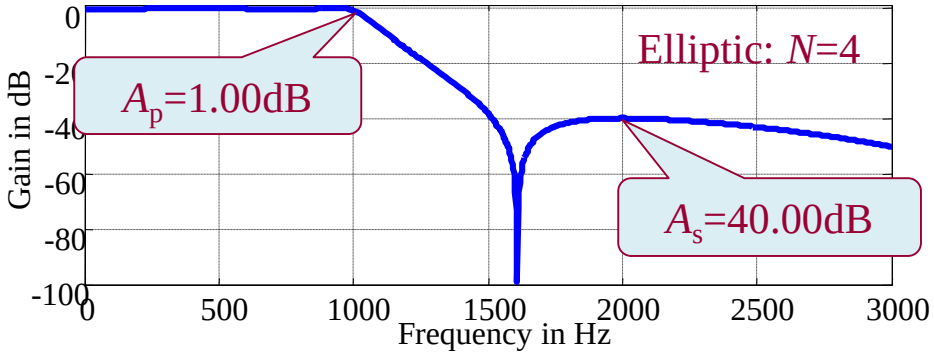
$$f_p=1\text{kHz},\ f_s=2\text{kHz},\ A_p\leq 1\text{dB},\ A_s\geq 40\text{dB}$$

Try to compare the order of three different filters。

Comparison of BW, CB and elliptic analog low-pass filters

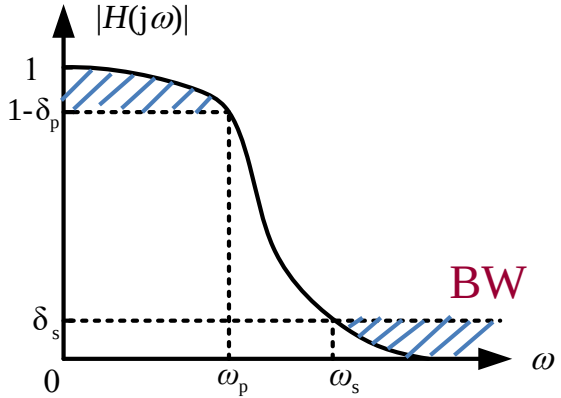


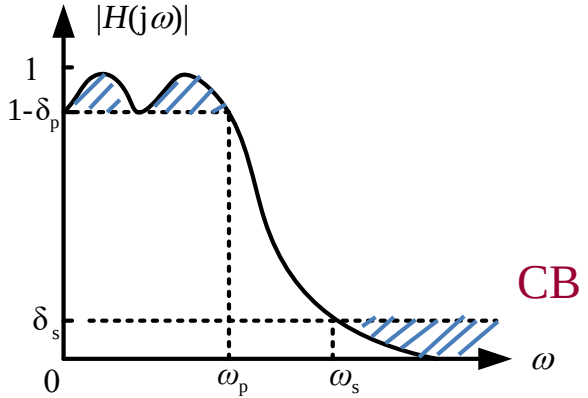


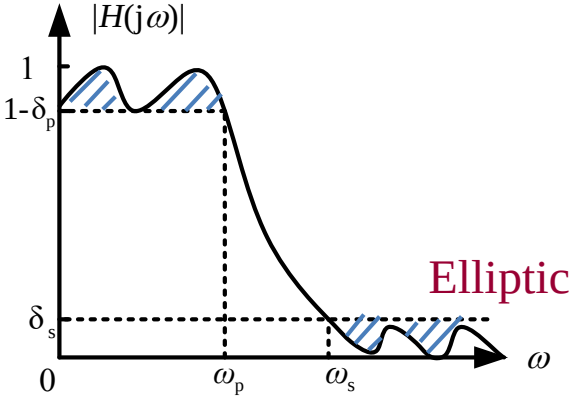


Conclusion: Under the same design index, the order of BW filter is generally the highest, and the order of elliptic filter is generally the lowest.

Comparison of BW, CB and elliptic analog low-pass filters





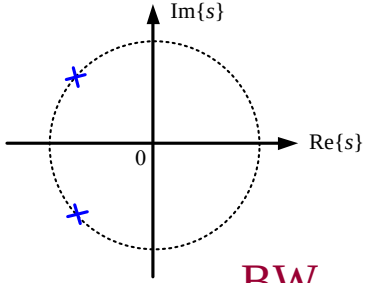


Under the same design index, the amplitude response of BW filter is monotone change, and the margin is the highest, so its order is generally the highest, while the elliptic filter has equal ripple characteristics in the stopband, and the margin is the least, so its order is generally the lowest。

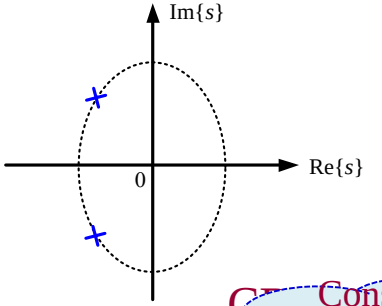
Comparison of BW, CB and elliptic analog low-pass filters

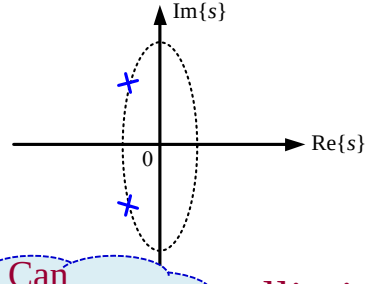
N is even

(N=2)



BW



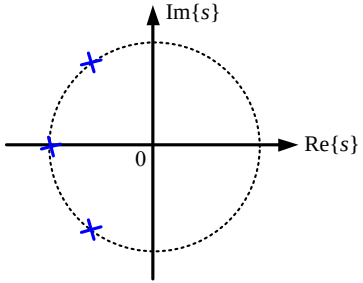


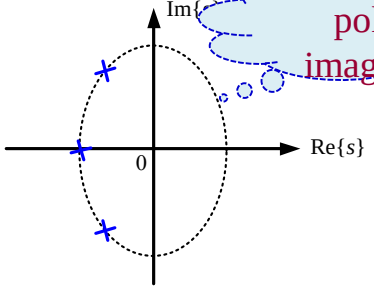
elliptic

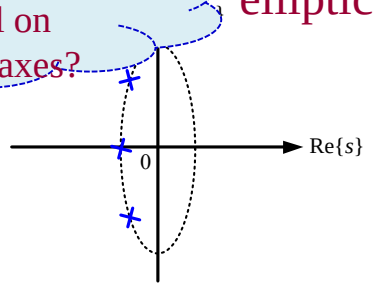
Consider: Can poles fall on imaginary axes?

N is *odd*

(N=3)







Conclusion: In the process of filter implementation, the BW filter is the easiest to implement, while the elliptic filter is not easy to implement (because the pole of the system function $H(s)$ is closest to the jω axis).

Analog low-pass filter design

BW type analog low-pass filter

```
[N,wc]=buttord(wp,ws,Ap,As,'s')
```

The order *N* and 3dB cut frequency wc of BW analog low-pass filter are determined. ω_c is determined by the stopband parameter. '*s*' indicates the design analog filter.

```
[num,den]=butter(N,wc,'s')
```

The numerator and denominator polynomial coefficients of BW analog low-pass filter with order *N* and 3dB cut frequency ω_c(radian/s) are determined. '*s*' indicates the design analog filter.

Analog low-pass filter design

Cb-i analog low-pass filter

```
[N,wc]=cheb1ord(wp,ws,Ap,As,'s')
```

The order *N* of CB I analog low-pass filter is determined. Where $\omega_c=\omega_p$(rad/s)

```
[num,den]=cheby1(N,Ap,wc,'s')
```

The CB-I analog low-pass filter with order *N* and passband intercept frequency ω_c(radian/s) is determined.Numerator and denominator polynomial coefficients. '*s*' indicates the design analog filter.

Analog low-pass filter design

CB II analog low-pass filter

```
[N,wc]=cheb2ord(wp,ws,Ap,As,'s')
```

The order *N* of CB II analog low-pass filter is determined.

```
[num,den]=cheby2(N,As,wc,'s')
```

The polynomial coefficients of numerator and denominator of CB II analog low-pass filter with order *N* and stopband attenuation A_s are determined.

Analog low-pass filter design

Elliptic analog low-pass filter

```
[N,wc]=ellipord(wp,ws,Ap,As,'s')
```

The order N of elliptic analog low-pass filter is determined. Where $\omega_c=\omega_p$(rad/s).

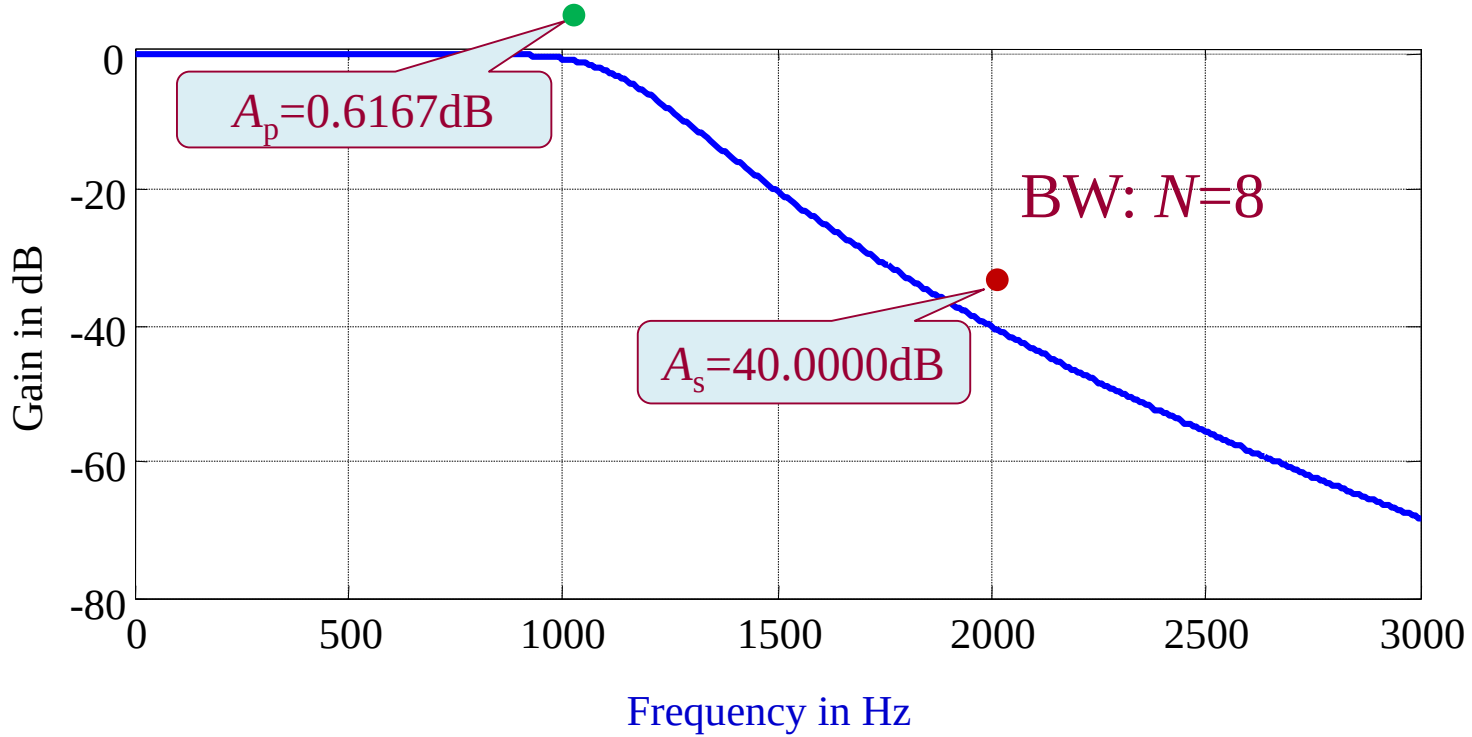
```
[num,den]=ellip(N,Ap,As,wc,'s')
```

The polynomial coefficients of numerator and denominator of elliptic analog low-pass filter with order N, passband attenuation A_p and stopband attenuation A_s are determined. ω_c is the passband truncation of an elliptic filter.

[Example] Design a BW type analog low-pass filter that meets the following criteria f_p=1kHz, f_s=2kHz, A_p≤1dB, A_s≥40dB。

```
Wp=2*pi*1000;Ws=2*pi*2000;Ap=1;As=40;   % Analog filter specifications
[N,Wc]=buttord(Wp,Ws,Ap,As,'s');    %Computer the order of analog filter
fprintf('Order of the filter=%.0f\n',N)
[num,den] = butter(N,Wc,'s');     %Compute AF coefficients
disp('Numerator polynomial'); fprintf('%.4e\n',num);
disp('Denominator polynomial'); fprintf('%.4e\n',den);
omega=[Wp Ws]; h = freqs(num,den,omega);   %Compute Ap and As of AF
fprintf('Ap= %.4f\n',-20*log10(abs(h(1))));
fprintf('As= %.4f\n',-20*log10(abs(h(2))));
omega = [0: 200: 3000*2*pi];
h = freqs(num,den,omega);  %Compute the frequency response of the AF
gain=20*log10(abs(h));  plot(omega/(2*pi),gain);
xlabel('Frequency in Hz');  ylabel('Gain in dB');
```

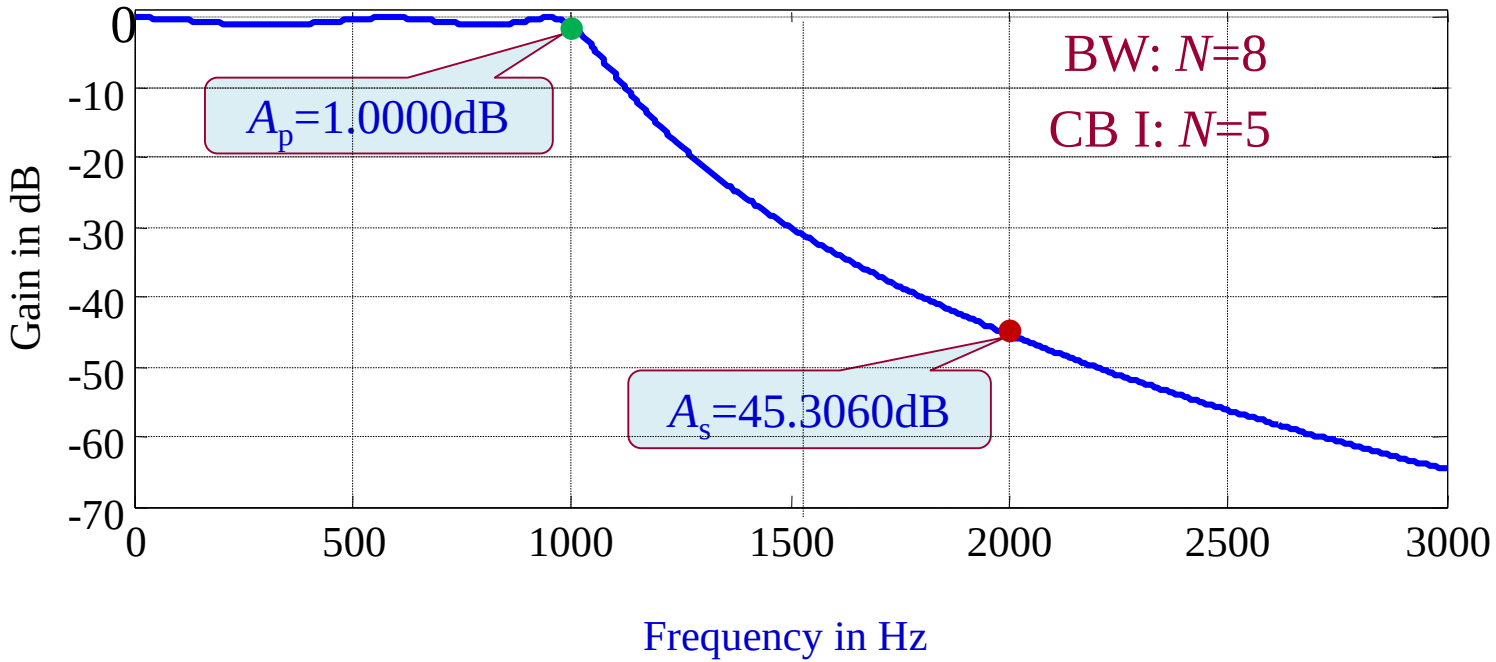
[Example] Design a BW type analog low-pass filter that meets the following criteria f_p=1kHz, f_s=2kHz, A_p≤1dB, A_s≥40dB。



[Example] Design a type CB I analog low-pass filter that meets the following criteria f_p=1kHz, f_s=2kHz, $A_p \leq$1dB, $A_s \geq$40dB

```
Wp=2*pi*1000;Ws=2*pi*2000;Ap=1;As=40;    % Analog filter specifications
[N,Wc]=cheb1ord(Wp,Ws,Ap,As,'s');  %Computer analog filter order
fprintf('Order of the filter=%.0f\n',N)
[num,den] = cheby1(N,Ap,Wc,'s');  %Compute AF coefficients
disp('Numerator polynomial'); fprintf('%.4e\n',num);
disp('Denominator polynomial'); fprintf('%.4e\n',den);
omega=[Wp Ws];  h = freqs(num,den,omega); %Compute Ap and As of the AF
fprintf('Ap= %.4f\n',-20*log10(abs(h(1))));
fprintf('As= %.4f\n',-20*log10(abs(h(2))));
omega = [0: 200: 3000*2*pi];
h = freqs(num,den,omega);  %Compute the frequency response of the AF
gain=20*log10(abs(h));  plot(omega/(2*pi),gain);
xlabel('Frequency in Hz');  ylabel('Gain in dB');
```

[Example] Design a type CB I analog low-pass filter that meets the following criteria f_p=1kHz, f_s=2kHz, $A_p \leq$1dB, $A_s \geq$40dB



Thank you

Some of the materials quoted in this course come from many years of teaching accumulation of the lecturer, from various media and the communication of colleagues and peers, it is difficult to specify the sources.